JOINs, UPDATE, DELETE, & more

COMP1048: Databases and Interfaces (2025-2026)

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1 Lab Overview

This week's lab builds on the previous one, using an extended version of the database. You will work with a pre-populated database and perform more advanced queries to answer the questions below. You can find the database file on Moodle. Write SQL queries for each question.

2 Exercises

Before starting, familiarise yourself with the database schema. Type `.schema` and press Enter to display the schema of the database.

1. Write a query that shows the movie title, year of release, and the leading actor for each movie. Include all movies, even if they don't have a leading actor. Arrange the output by year of release, first in ascending order, and then by movie title in ascending order.

year	movie_title	leading_actor
...	...	...

2. Write a query that shows the total sales for each region (for all movies). Arrange the output by region name in ascending order. Your result should appear as follows:

region	total_sales
...	...

3. Write a query that shows the sales number and region for all movies. If a movie has sales data from multiple regions, each movie-region pairing should appear on a separate row. Include all movies in the output at least once, even if they have no sales data. Arrange the output by movie title in ascending order. Your result should appear as follows:

movie_title	region	sales
...	...	...

4. Write a query that shows the total sales number (summed over all regions) for each movie. Arrange the output by sales, in descending order. Include all movies, even if they have no sales data. Your result should appear as follows:

movie_title	sales
...	...

5. Write a query that shows the total sales for all movies (across all regions) that an actor has starred in. Include all actors, even if they have not appeared in any films. Arrange the output by sales, in descending order. Your result should appear as follows:

actor_name	sales
...	...

6. Write a query that increases the sales number of all movies released in Europe by 10%, rounded to the nearest integer. Your query should produce no output.

7. Write a query that shows the income for each movie. The income is the total sales number (summed over all regions) multiplied by the movie price, rounded to two decimal places. Arrange the output by year of release, first in ascending order, then by movie title in ascending order. You do not need to include movies with no sales in your result. Your result should appear as follows:

year	movie_title	income
...	...	...

8. Write a query that determines the most popular actor in each region. The actor's popularity in one region is measured by the **average sales** of all movies in that region that the actor has appeared in. Arrange the output by region name in ascending order. Your result should appear as follows:

region	actor_name	average_sales
...	...	...

> **Hint**
>
> In SQL, `GROUP BY` can be used to group rows that have the same value in more than one column. For example, the following query will group all rows that have the same value in the `year` and `title` columns.
>
> ```sql
> SELECT year, title, SUM(sales) AS total_sales
> FROM movies
> GROUP BY year, title;
> ```

9. Write a query that removes all movies without any sales data. The output should not show anything.

3 Submitting Your Lab Work

This week, as with last week, you will submit your lab work through a Moodle quiz. For each question, submit an SQL statement via the quiz. You'll receive immediate feedback on each submission (correct or incorrect) and can resubmit as many times as needed before the deadline, without penalty.

Submit all answers before the deadline, as late submissions will not be accepted, and a mark of zero will be recorded for the lab. Unlike other labs, your grade for this lab will be based on the number of correct answers submitted, so it's important to attempt all questions and resubmit until they are correct. There is no penalty for multiple attempts.

This lab contributes 1% of your overall module grade.